



Daffodil
International
University

ASSIGNMENT

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Course Title: COMPUTER NETWORK

Topic Name: VoIP and Unified Communications Network
Design for a Corporate Office

Daffodil International University

Submitted To

Name: Safayat Hossain

Designation: Lecturer

Department of Computing & Information System

Daffodil International University

Submitted By

Name: ABU BOKKOR RAHMAN , AKIBUL HASSAN ,
MOLOY ROY , MUHAJMINUL SHAMMO

ID: 252-16-036 , 252-16-037 , 252-16-022 , 252-16-007

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Department of CIS

Daffodil International University

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Question 1: OSI Model Application, Network Architecture, and Topology

1. Network Type and Architecture

Network Type: Local Area Network (LAN)

A **Local Area Network (LAN)** is the optimal choice for a corporate office environment deploying **Voice over IP (VoIP)** and **Unified Communications (UC)**. A LAN facilitates high-speed, low-latency data exchange between IP phones, workstations, and servers within a single physical location. For broader organizational needs, multiple LANs can be interconnected via a **Wide Area Network (WAN)**.

Network Architecture: Client-Server

The network utilizes a **Client-Server** architecture. In this model, end-user devices such as IP phones serve as clients, while a central **IP Private Branch Exchange (PBX)** or UC Server manages core functions, including call processing, authentication, and directory services.

Core Principle:

- Centralized administration of users and communication services.
- Enhanced security through unified authentication protocols.
- Streamlined maintenance and software deployment.
- Scalable design accommodating future organizational growth.
- Robust resource management and high system availability.

2. Network Topology

Selected Topology: Extended Star Topology

An **Extended Star Topology** is highly recommended for modern corporate infrastructures. In this configuration, a central **core switch** connects to several

access switches, which in turn connect to end devices like IP phones and computers.

In this design, a **Core Switch** is connected to multiple **Access Switches**, and each access switch connects end devices such as IP phones, computers, printers, and wireless access points.

Key Insight:

- High fault tolerance; individual cable failures do not disrupt the entire network.
- Simplified troubleshooting and rapid fault detection.
- Seamless expansion capabilities without requiring architectural overhauls.
- Optimized performance for medium-to-large enterprise environments.
- Consistent reliability across all connected nodes.

3. Application of the OSI Reference Model in VoIP Communication

The **OSI model** explains how voice data is transmitted across the network during a VoIP call.

Conclusion

For a VoIP and Unified Communications network, a **LAN-based Client-Server Architecture** with an **Extended Star Topology** is the most effective solution. It provides high-speed communication, centralized management, scalability, reliability, and enhanced security, making it ideal for a modern corporate office.

Question 2: Critical Network Protocols and Their Roles

VoIP and UC networks operate through the integration of protocols across multiple OSI/TCP-IP layers. Three foundational protocols for this scenario are analyzed below.

OSI Layer	Application in VoIP Communication
Layer 7 – Application	SIP (Session Initiation Protocol) establishes, manages, and terminates voice and video calls.
Layer 6 – Presentation	Converts voice data into a suitable format using audio codecs such as G.711 or G.729 and may perform encryption.
Layer 5 – Session	Establishes, maintains, and terminates communication sessions between users.
Layer 4 – Transport	UDP is commonly used to deliver real-time voice packets with low latency, while TCP may be used for signaling when required.
Layer 3 – Network	IP addresses are used to route voice packets from the source to the destination across the network.
Layer 2 – Data Link	Ethernet switches use MAC addresses to forward frames and can separate voice traffic using Voice VLANs .
Layer 1 – Physical	Voice signals are transmitted through physical media such as Cat6 Ethernet cables or fiber-optic cables.

1. Session Initiation Protocol (SIP) – Application Layer

SIP is an application-layer signaling protocol essential for initializing, managing, and terminating multimedia sessions, including voice and video conferencing.

Core Principle:

- Establishes and terminates VoIP call sessions.
- Registers endpoints with the central IP PBX.

- Supports advanced features like presence and instant messaging.

2. User Datagram Protocol (UDP) – Transport Layer

UDP is a connectionless transport-layer protocol optimized for speed and low overhead, making it ideal for real-time applications where latency is a primary concern.

Key Insight:

- Facilitates low-latency transport of voice packets.
- Eliminates delays associated with connection handshakes and retransmissions.
- Ensures smooth, jitter-free voice delivery during active calls.

3. Internet Protocol (IP) – Network Layer

IP is a fundamental network-layer protocol responsible for the logical addressing and routing of packets across diverse network segments.

Core Principle:

- Manages source and destination addressing for all voice data.
- Routes traffic efficiently between departments and remote sites.
- Enables cross-platform communication between devices and servers.

Conclusion

SIP, **UDP**, and **IP** work together to provide efficient VoIP communication. SIP establishes and manages calls, UDP delivers voice data with low latency, and IP routes voice packets to their correct destinations. Together, these protocols ensure reliable, real-time communication within the corporate office network.

Task 2: Applying IP Addressing, Subnetting, VLAN, and Routing

Question 3: IP Addressing, Subnetting, and VLAN Segmentation

IP Addressing Scheme

To ensure security and traffic isolation, the network is partitioned into four primary subnets: IT, Sales, HR, and Voice. A **/24 subnetting strategy** is applied to the **192.168.0.0/16** private address block to provide ample host capacity and simplified management.

Department / VLAN	Network Address	Prefix	Subnet Mask	Default Gateway	Usable Host Range	Broadcast Address
IT (VLAN 10)	192.168.10.0	/24	255.255.255.0	192.168.10.1	192.168.10.2 – .254	192.168.10.255
Sales (VLAN 20)	192.168.20.0	/24	255.255.255.0	192.168.20.1	192.168.20.2 – .254	192.168.20.255
HR (VLAN 30)	192.168.30.0	/24	255.255.255.0	192.168.30.1	192.168.30.2 – .254	192.168.30.255
Voice (VLAN 100)	192.168.100.0	/24	255.255.255.0	192.168.100.1	192.168.100.2 – .254	192.168.100.255

VLAN ID	Department	Purpose
VLAN 10	IT Department	Data traffic for IT staff
VLAN 20	Sales Department	Data traffic for Sales staff

VLAN ID	Department	Purpose
VLAN 30	HR Department	Data traffic for HR staff
VLAN 100	Voice VLAN	IP Phones and VoIP communication

VLAN Segmentation Justification

The network is divided into separate **VLANs** to improve **security, performance, and manageability**.

- Security: Isolates sensitive departmental data within distinct broadcast domains.
- Performance: A dedicated Voice VLAN ensures voice traffic is prioritized via QoS.
- Reliability: Reduces congestion by limiting the scope of broadcast traffic.
- Scalability: Allows for the addition of new subnets without hardware reconfiguration.

Question 4: Routing Protocol Application and Implementation

Comparison of RIP and OSPF

Feature	RIP	OSPF
Routing Type	Distance Vector	Link State
Metric	Hop Count	Cost (Bandwidth)
Max Hops	15	Unlimited
Convergence	Slow	Fast
Suitability	Small networks	Large networks

Selection: OSPF (Open Shortest Path First)

OSPF is the most suitable routing protocol for the proposed corporate office network.

OSPF is selected over **RIP** due to its superior convergence speed and scalability. For a VoIP environment, OSPF's ability to rapidly recalculate paths in the event of a link failure is critical to maintaining call quality and preventing session drops.

OSPF Configuration Decisions

The following configuration decisions will be applied:

- Deployment of all routers within a single Backbone Area (Area 0).
- Implementation of Inter-VLAN routing via Layer 3 switches.
- Enforcement of MD5 authentication for secure routing updates.
- Configuration of passive interfaces on edge ports to limit overhead.

Justification

OSPF is the preferred routing protocol because it provides fast convergence, high scalability, efficient bandwidth utilization, and reliable routing. These features ensure uninterrupted voice communication, reduced latency, and excellent performance for VoIP and Unified Communications applications. Therefore, OSPF is the best choice for a modern corporate office network.

Task 3: Analyzing Application and Network Protocols, Media, and Hardware

Question 5: Application-Layer Protocol Analysis

Two essential protocols for supporting the VoIP infrastructure are **DHCP** and **HTTPS**. DHCP ensures efficient network onboarding, while HTTPS provides the necessary security for administrative interfaces.

1. Dynamic Host Configuration Protocol (DHCP)

Definition:

DHCP is an application-layer protocol that automatically assigns IP addresses and other network configuration parameters to network devices.

Use Case in the Corporate Office

- Automatically assigns IP addresses to IP phones, computers, and other network devices.
- Provides subnet mask, default gateway, DNS server, and Voice VLAN information.
- Supplies the TFTP server address used by IP phones to download their configuration files.

2. Hypertext Transfer Protocol Secure (HTTPS)

Definition:

HTTPS is the secure version of HTTP and uses **SSL/TLS** encryption to provide secure communication between users and web servers.

Use Case in the Corporate Office

- Provides secure access to the IP PBX or Unified Communications Server management portal.
- Allows users to securely access voicemail, call history, and web-based communication services.
- Protects administrator login credentials and sensitive communication data.

Feature	DHCP	HTTPS
OSI Layer	Application	Application
Transport	UDP (Ports 67 & 68)	TCP (Port 443)
Reliability	Best-effort	Reliable (TCP)

Conclusion

DHCP simplifies network administration by automatically configuring network devices, while **HTTPS** provides secure communication and protects sensitive management data. Together, these protocols improve the efficiency, security, and reliability of a VoIP and Unified Communications network.